

Impact of pharmacy involvement on care gap closure in Managed Medicaid patients

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Purpose: A population health pharmacy team was developed to improve health and increase reimbursement for patients with value-based care plans. The purpose of this study was to evaluate the impact of pharmacy involvement on care gap closure in Managed Medicaid patients.

Methods: This was a single-center, retrospective cohort study of Managed Medicaid patients seen at outpatient facilities. Patients had a pharmacy risk score (PRS) of 6 or greater and had not achieved the Medicaid quality measures for both glycated hemoglobin (HbA_{1c}) and blood pressure (BP). The intervention group included patients reviewed by pharmacy, compared to a control group of patients not reviewed by pharmacy. The primary outcome was closure of at least one care gap by the end of 2022. Secondary outcomes were the number of each type of gap closed, the frequency at which recommendations were made by pharmacists, and the frequency at which pharmacist recommendations were implemented by providers.

Results: Data were collected for 80 patients for the period from January through October 2022. The primary outcome occurred in 37 (74%) patients in the intervention group and 15 (50%) patients in the control group (odds ratio, 2.85; $P = 0.032$). The HbA_{1c} gap was closed in 30 (60%) patients in the intervention group and 8 (27%) patients in the control group. The BP gap was closed in 24 (48%) patients in the intervention group and 11 (37%) patients in the control group. The frequency with which recommendations were made by a pharmacist was associated with gap closure ($P = 0.012$). No significant difference was found based on the frequency at which recommendations were implemented by providers ($P = 0.4$).

Conclusion: Pharmacy intervention was associated with an almost 3-fold-higher likelihood of closing at least one care gap in Medicaid patients. HbA_{1c} gap closure was achieved more frequently than BP gap closure due to pharmacy involvement. The frequency with which recommendations were made by pharmacy was associated with increased gap closure regardless of the frequency with which these recommendations were implemented by providers.

Keywords: diabetes, Medicaid, patient outreach, pharmacy intervention, population health, risk stratification

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The Centers for Medicare and Medicaid Services (CMS) are moving away from the traditional fee-for-service healthcare model to value-based care (VBC) reimbursement models. With VBC, providers and healthcare institutions are reimbursed based on annual patient outcomes as opposed to being paid per service they deliver. The goal of

VBC is to improve society's overall health while decreasing healthcare costs.¹ Aligning with the goal of implementing VBC, Medicaid created Medicaid Managed Care, or "Managed Medicaid," a healthcare delivery system that contracts through managed care organizations (MCOs) and prepaid health plans (PHPs) to optimize quality and minimize

costs of services.² MCOs and PHPs provide comprehensive care to Medicaid beneficiaries, accepting capitated (per member per month) payment. Medicare and several commercial insurance plans operate via VBC as well.

As of February 2023, over 2.3 million individuals were enrolled in Medicaid in North Carolina, corresponding to nearly one-fifth of the state's population.² NC Medicaid adopted Managed Medicaid on July 1, 2021.³ NC Managed Medicaid contracts with PHPs, which include the standard plans introduced in 2021 (AmeriHealth Caritas, Carolina Complete Health, Healthy Blue, UnitedHealthcare Community Plan, and WellCare).⁴ To standardize data collection, reporting, and reimbursement, CMS developed quality measures, or care gaps, intended to serve as indicators for the assessment of quality care provided to Managed Medicaid beneficiaries under VBC models.⁵ With such high risk-reward stakes, all members of the healthcare team must be involved to best care for patients and receive full reimbursement for patients managed. Several of the Medicaid care gaps can be directly influenced by medication management, including "comprehensive diabetes care: hemoglobin A1c poor control," which patients are passing if their glycated hemoglobin (HbA_{1c}) level is less than 9%, and "controlling high blood pressure," which patients are passing if their systolic blood pressure (SBP) is less than 140 mm Hg and their diastolic blood pressure (DBP) is less than 90 mm Hg.⁶ Pharmacy teams are well positioned to impact these medication-related quality measures and increase Managed Medicaid reimbursement. **The aim of this project was to determine whether pharmacy involvement leads to increased closure of care gaps for diabetes control and hypertension in Managed Medicaid patients.**

Methods

Pharmacy population health team. The organization joined an accountable care organization and began



Ellen Montgomery is the current postgraduate year 2 internal medicine pharmacy resident at Atrium Health Wake Forest Baptist, where she also completed her postgraduate year 1 pharmacy residency. Ellen is a native of North Carolina, attending UNC Eshelman School of Pharmacy and Elon University before residency. Throughout residency training, Ellen was fueled by a passion for transitions of care and equal access to healthcare. Through direct patient care and research experiences, she gained an appreciation for meeting patients where they are and creatively utilizing tools in place to reach a more diverse patient population. Ellen looks forward to continuing to bridge the gap between inpatient care and chronic disease management through a role she has accepted after residency as a population health and acute care medicine (hybrid) pharmacist at her current institution. Outside of pharmacy, Ellen is learning to love running, completing her first half-marathon with coresidents and friends this past year.

participating in VBC in 2015.⁷ As of the beginning of 2022, the institution held 11 VBC contracts with over 150,000 covered individuals. In 2016, the Pharmacy Population Health Center (PPHC) was implemented to manage patients covered by VBC plans in the outpatient setting with the vision of "providing medication-related strategies designed to optimize patient care and outcomes, improve health, and reduce the total cost of care, while

considering determinants of health." PPHC includes pharmacists embedded within income-based clinics as well as decentralized pharmacists and pharmacy technicians. The decentralized teams work within designated "service lines" for the various VBC contracts.

As part of the PPHC workflow, a **pharmacy risk score (PRS)** was developed to identify patients within large VBC patient populations who could benefit from pharmacy involvement. This metric was originally developed for the traditional Medicare population and then expanded to other VBC plans. The PRS is calculated in the electronic health record (EHR) and stratifies patients based on risk factors that may lead them to fail care gaps (**Figure 1**). Reports can be generated for patients in different VBC populations and sorted by PRS to aid the pharmacy team in prioritizing patients to review and contact if necessary. This tool won a Best Practices Award from the American Society of Health-System Pharmacists (ASHP) in 2019.⁷

The Medicaid workflow for the population health pharmacy team utilized the PRS to identify patients to target for intervention, with the goal of increasing gap closure. **A report was generated weekly** of Managed Medicaid patients attributed to the health system who were seen for scheduled appointments at outpatient clinics the previous week. Patients were then filtered and sorted by HbA_{1c}, BP, and PRS from high to low. Patients with a PRS of 6 or greater (considered medium or high risk) were pulled for review by the pharmacy team. Included patients were first screened by certified pharmacy technicians, who reached out to patients and assessed barriers to care before escalating encounters to a pharmacist. The pharmacist performed an in-depth chart review, focusing on disease states with outcomes directly linked to Medicaid care gaps (diabetes and hypertension), as well as general drug-related problems. The pharmacist then contacted the patient again to gather more information, provide medication counseling, and

Figure 1. Components and interpretation of the pharmacy risk score (PRS). COPD indicates chronic obstructive pulmonary disease; CVD, cardiovascular disease; HbA_{1c}, glycated hemoglobin.

Components for Calculating PRS		Interpreting Score							
Patient age	Has asthma	<table><tr><td>Low Risk</td><td>0-5</td></tr><tr><td>Medium Risk</td><td>6-15</td></tr><tr><td>High Risk</td><td>16-23</td></tr></table>		Low Risk	0-5	Medium Risk	6-15	High Risk	16-23
Low Risk	0-5								
Medium Risk	6-15								
High Risk	16-23								
More than 2 hospital admissions	High-risk CVD and no statin								
Discharge in past 30 days	Renal or hepatic impairment								
Uncontrolled hypertension	Chronic opioid use								
Diabetes HbA _{1c} > 8%	No current Narcan prescription								
Has heart failure	High-cost medication								
Has COPD	Composite medication count								
Has behavioral health diagnosis	Social determinants of health								

make recommendations if necessary. Pharmacotherapy recommendations were documented in a pharmacy note as well as sent directly in a message to the patient's primary care provider (PCP) in the EHR.

Study design and patient population.

This was a single-center, retrospective cohort study using data from January through October 2022 for patients with a Managed Medicaid insurance plan attributed to a large academic health system. One hundred patients were randomly selected from a population of over 1,500 patients using simple randomization. Patients were pulled from weekly reports of Managed Medicaid patients seen at outpatient clinics by a PCP (either in person or via telehealth) within the health system. The study population included patients 18 years of age or older with a PRS greater than or equal to 6 and both an elevated HbA_{1c} level (greater than 9%) and BP (SBP greater than 140 mm Hg or DBP greater than 90 mm Hg) who were seen at outpatient clinics at least twice in 2022. Patients were excluded if their HbA_{1c} level was less than 9%, their SBP was less than 140 mm Hg, and their DBP was less than 90 mm Hg or if they lacked repeat HbA_{1c} and BP values following the initial clinic visit. The intervention group comprised patients who were reviewed by the pharmacy team, compared to a control group of patients who were not reviewed by the pharmacy team due to the high volume of patients attributed to the

health system and a limited number of pharmacists.

Data collection. Data were collected via chart review of the EHR. Demographics were collected, including medical record number, sex, age, race, region attributed to, and PHP. Baseline values of PRS, HbA_{1c}, SBP, and DBP were collected. These data points were collected again at 3 months and 6 months (if applicable), as well as whether the patient was passing either Medicaid care gap. In the intervention group, the number of pharmacy encounters in the specified timeframe was collected, along with the frequency of recommendations made by pharmacists (documented in pharmacy notes in the EHR) and frequency of recommendations implemented by PCPs (determined based on changes in medication regimens after a recommendation was documented).

Outcomes. The primary outcome was closure of at least one care gap (either "comprehensive diabetes care" or "controlling high blood pressure") as determined using the last documented value in the EHR in 2022. Secondary outcomes included the number of each type of gap closed, the frequency with which recommendations were made by pharmacists, and the frequency with which recommendations were implemented by providers.

Statistical analyses. Baseline characteristics were analyzed using χ^2 , Fisher's exact, and Wilcoxon rank-sum tests. Univariate statistics and χ^2

tests were used to evaluate the primary outcome. Secondary outcomes were evaluated using descriptive statistics. The study was powered at 80% to detect a 50% difference in gap closure with an α value of 0.05.

Results

Patient characteristics. A total of 80 patients were included in the study population, with 50 in the intervention group and 30 in the control group (Table 1). There was no difference between the groups in median age ($P = 0.10$), sex ($P = 0.13$), race ($P > 0.9$), HbA_{1c} ($P = 0.9$), SBP ($P = 0.9$), or DBP ($P = 0.7$). A significant difference was detected in PRS at baseline, with a median of 8 in the intervention group and 7 in the control group ($P = 0.002$). PHP types were also collected, with no difference found between the groups.

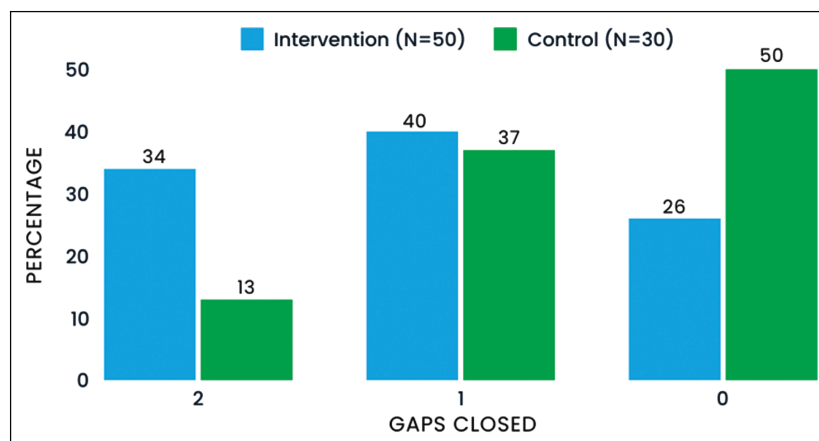
Primary outcome. The primary outcome, which was closure of at least one care gap, occurred in 37 (74%) patients in the intervention group and 15 (50%) patients in the control group (odds ratio, 2.85; 95% confidence interval, 1.10-7.54; $P = 0.032$). The percentage of patients in each group with 2, 1, and 0 gaps closed was also assessed (Figure 2). Two gaps were closed in 34% of patients in the intervention group compared to 13% of patients in the control group. One gap was closed in 40% of patients in the intervention group compared to 37% of patients in the control group. No gaps were closed in 26% of patients in the intervention

Table 1. Baseline Characteristics

Characteristic ^a	Intervention group (n = 50)	Control group (n = 30)	P value
Age, median (IQR), years	56 (45-61)	48 (38-57)	0.10
Sex			0.13
Female	32 (64)	24 (80)	
Race			>0.9
White	24 (48)	14 (48)	
Black	24 (48)	14 (48)	
Other	2 (4)	2 (6.8)	
PRS, median (IQR)	8 (7-9)	7 (6-8)	0.002
HbA _{1c} , median (IQR), %	11.1 (9.7-12.1)	10.8 (9.8-12.7)	0.9
SBP, median (IQR), mm Hg	151 (144-165)	150 (148-159)	0.9
DBP, median (IQR), mm Hg	88 (81-94)	90 (76-101)	0.7

Abbreviations: DBP, diastolic blood pressure; HbA_{1c}, glycated hemoglobin; IQR, interquartile range; PRS, pharmacy risk score; SBP, systolic blood pressure.

^aData shown as No. (%) unless indicated otherwise.

Figure 2. Percentage of patients in both the intervention and control groups with 2, 1, and 0 gaps closed by the end of the calendar year.

group compared to 50% of patients in the control group.

Secondary outcomes. The number of each type of care gap closed was evaluated. The “comprehensive diabetes care” gap was closed in 30 (60%) patients in the intervention group compared to 8 (27%) patients in the control group. The “controlling high blood pressure” gap was closed in 24 (48%) patients in the intervention group compared to 11 (37%) patients in the control group. In the intervention group, follow-up values for HbA_{1c}, SBP, and DBP were collected at least 3

months from baseline and following pharmacy involvement (Table 2).

The rate of recommendations made by pharmacists to providers was assessed as well as the rate of implementation of recommendations by providers (Table 3). This outcome was only assessed in the intervention group, where the 37 patients who met the primary outcome were compared to the 13 patients who did not. The median number of recommendations made by a pharmacist in patients who met the primary outcome was 1, compared to 0 in those who did not ($P = 0.012$). No

difference was found between patients in the intervention group who did and did not meet the primary outcome in the frequency of recommendations implemented by providers (median of 0 in both groups; $P = 0.4$).

Discussion

Interdisciplinary collaboration and longitudinal management at the population level for patients with VBC plans are integral to achieving gap closure and receiving maximum reimbursement for patients covered. This retrospective cohort study of Managed

Table 2. Follow-Up Glycated Hemoglobin and Blood Pressure Measures^a

Measure	Intervention group (n = 50)	Control group (n = 30)
HbA _{1c} , median (IQR), %	8.9 (7.7-9.5)	10.5 (8.9-11.7)
SBP, median (IQR), mm Hg	142 (123-159)	143 (132-150)
DBP, median (IQR), mm Hg	82 (73-91)	83 (76-89)

Abbreviations: DBP, diastolic blood pressure; HbA_{1c}, glycated hemoglobin; IQR, interquartile range; SBP systolic blood pressure.
^aRepeat laboratory and vital values were collected at least 3 months from baseline.

Table 3. Rate of Recommendations Made by Pharmacists and Implemented by Providers in the Intervention Group

Recommendation type ^a	Met primary outcome (n = 37)	Did not meet primary outcome (n = 13)	P value
Made by a pharmacist	1 (0-3)	0 (0-2)	0.012
Implemented by a provider	0 (0-3)	0 (0-1)	0.4

^aData shown as median (interquartile range) unless indicated otherwise.

Medicaid patients attributed to a large academic health system directly assessed the impact of pharmacy involvement on gap closure in patients at moderate to high risk based on PRS. The primary outcome, which was closure of at least one care gap by the end of 2022, demonstrated that patients who were reviewed by the pharmacy team were almost 3 times more likely to achieve gap closure than those who were not reviewed by pharmacy, based on an odds ratio of 2.85. This significant improvement in care with the population health pharmacy team should be a point of emphasis for health systems when evaluating ways to optimize reimbursement for VBC patient populations.

When looking at secondary outcomes, the “comprehensive diabetes care” gap was closed more frequently than the “controlling high blood pressure” gap, indicating that it may be more feasible to reduce HbA_{1c} than to lower BP through pharmacy involvement, although this is difficult to assess as the number of recommendations related to each disease state was not assessed. Diabetes management could be an effective initial point of focus for population health pharmacy teams to see the greatest benefit. Additionally, the frequency of recommendations made by a pharmacist was associated

with improved gap closure, while the frequency of recommendations implemented by providers was not. These results suggest that simply drawing attention to patients who were not achieving care gaps increased gap closure and improved overall health.

The PRS ranks patients based on risk factors that would lead to failure of gap closure in a population of thousands of VBC patients. While the PRS won an ASHP Best Practices Award and has been utilized in the workflow, this was the first study evaluating it in prioritizing appropriate patients for pharmacy review. The significant results for the primary outcome in a targeted population of patients with a PRS of 6 or greater suggest that the PRS is a useful tool for identifying patients who would benefit from pharmacy involvement, although further research is needed to validate the tool’s accuracy. Additionally, the results demonstrate the value of utilizing a risk score when determining which patients the pharmacy team should focus on to optimize gap closure in a large population of patients.

Limitations and future research. There were several limitations to this study, including a decrease in power due to fewer patients meeting the inclusion criteria in the control

group than anticipated. Despite a large population, only 30 patients were identified who did not receive the intervention and met all inclusion criteria, without meeting exclusion criteria. With 50 patients in the intervention group and 30 patients in the control group, the study was powered at 62% to detect a 50% difference with an α value of 0.05. Patients were included only if they were failing both the “comprehensive diabetes care” and “controlling high blood pressure” gaps. Including patients who were failing only one gap could have led to a more robust analysis of the impact of pharmacy involvement on individual gap closure. Additionally, nonpharmacological recommendations made by pharmacists were not captured in data collection. Lifestyle modifications, such as improved diet and increased aerobic exercise, are frequently recommended by pharmacists and can have a major impact on disease states such as diabetes and hypertension. By excluding these interventions, the results underestimate the impact of the pharmacy team on gap closure. Pharmacists involved in this study did not have a collaborative practice agreement, which differs from much of the preexisting literature on pharmacist medication optimization.

The intervention and control groups were not intentionally matched based on baseline characteristics, including PRS. This strategy for identification of the control and intervention groups could have led to the difference in baseline PRS and thus the resultant difference in gap closure and impact of pharmacist intervention. A major analytical limitation was the absence of a multivariable analysis, meaning that there was no control for other potential explanatory factors. Therefore, the study design was unable to demonstrate causation.

The findings of this study demonstrate the significant impact of pharmacy involvement on care gap closure and highlight areas for future research to further increase evidence for the value of pharmacy population health teams. A financial analysis of Managed Medicaid reimbursement for patients with pharmacy involvement would support continued growth of population health pharmacy teams. While this study validated the PRS for identifying patients at risk for gap failure, a score of 6 was arbitrarily selected as the breakpoint for moderate risk. A comparison of gap closure in patients with a PRS of less than 6 to the gap closure in patients included in this study would be useful to evaluate what PRS value truly indicates moderate to high risk.

Conclusion

Pharmacy involvement in Managed Medicaid patients via chart review, patient outreach, and recommendations made to providers was significantly associated with increased care gap closure. Health systems with large populations of VBC patients should consider utilizing pharmacy population health teams to maximize reimbursement through increased care gap closure and improve management of disease states such as diabetes and hypertension.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Disclosures

The authors have declared no potential conflicts of interest.

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